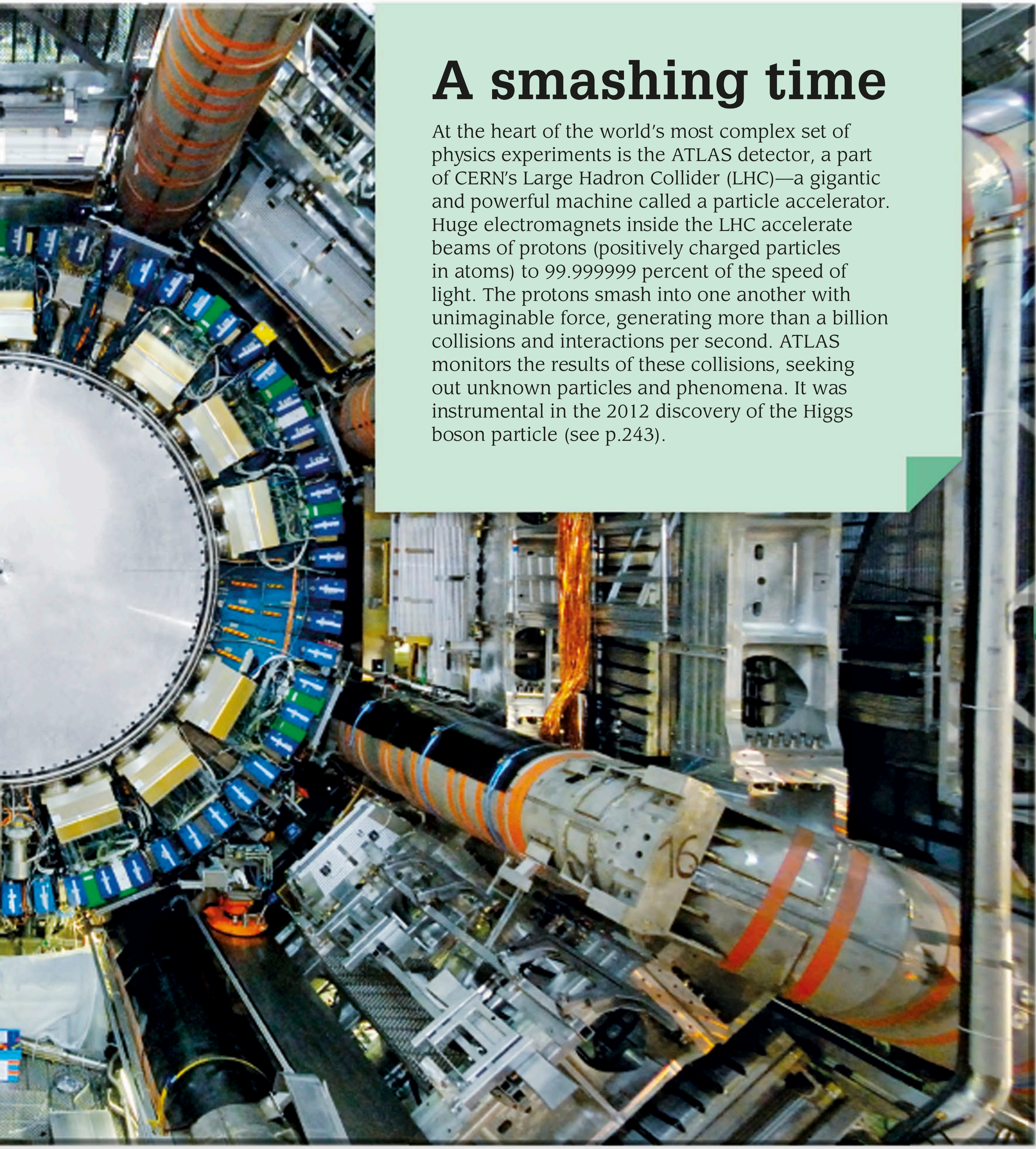
A smashing time

At the heart of the world's most complex set of physics experiments is the ATLAS detector, a part of CERN's Large Hadron Collider (LHC)—a gigantic and powerful machine called a particle accelerator. Huge electromagnets inside the LHC accelerate beams of protons (positively charged particles in atoms) to 99.999999 percent of the speed of light. The protons smash into one another with unimaginable force, generating more than a billion collisions and interactions per second. ATLAS monitors the results of these collisions, seeking out unknown particles and phenomena. It was instrumental in the 2012 discovery of the Higgs boson particle (see p.243).



The ATLAS detector is 82 ft (25 m) in diameter and weighs nearly 7,700 tons (7,000 metric tons), almost as much as the Eiffel Tower.